

Ascent sequences with bounded multiplicity (OEIS A294220)

Factorial growth constants of every cap $b \geq 3$, and fixed Taylor sectors of the large-cap expansion

A merged research report built from two manuscripts
(batch 77, manuscripts 47 and 52)

Manuscripts dated 1 October 2026; merged 2 October 2026

Abstract

Let $a_n^{(b)}$ count ascent sequences of length n in which each value occurs at most b times (the array OEIS A294220; cap three is A317784, cap two A202058). This report prints two manuscripts as two Parts. Part I proves, for every fixed $b \geq 3$, that $(a_n^{(b)}/n!)^{1/n} \rightarrow \mu_b = 1/T_b$ with $T_b = \int_0^\infty \log E_b(v)/(E_b(v) - 1) dv$ and E_b the truncated exponential, with a coarse error $O_b(n/\log n)$ in $\log a_n^{(b)}$, a controlled Lambert- W inverse in the length, the large-cap expansion $T_b = \pi^2/6 + (b+1)(\zeta(b+2) - 1) + O(b^{5/2}(4/9)^b)$ with explicit constants, and a rounding-safe inverse in the cap with counterexamples to naive rounding; the cap-two input $T_2 = 3\pi^2/8$ is taken from the sibling report on A202058. Part II writes $T_b - \pi^2/6$ exactly as a convergent sum of positive Taylor sectors $A_{r,b}$, proves that the remainder after any fixed number K of sectors is asymptotic to sector $K+1$, and expands every fixed sector to all algebraic orders in $1/b$; in particular the remainder of Part I is $\sim \frac{8}{9\sqrt{\pi}}\sqrt{b}(4/9)^b$. No prefactor, ratio limit or all-orders expansion in n for $b \geq 3$, no joint regime $b = b(n)$, and no uniformity in a growing number of sectors is claimed. The report is unrefereed, and none of it is formalized.

Contents

Guide to this report	2
I Factorial growth of ascent sequences with bounded multiplicity	7
1 Results and interpretation	7
2 Exact compaction and the counting operator	8
3 Characteristic curve and a uniform barrier	10
3.1 Exact child rank and its bounds	11
3.2 Frozen integral and residual	11
4 Comparison on the infinite state space	12
5 The radius and the coefficient limit	12
5.1 A concentration estimate from barrier values	13
5.2 Extraction for each sufficiently large length	13
5.3 A quantitative logarithmic error	14

6	Inverse in the sequence length	14
7	Large caps and an exact first variation	15
7.1	Evaluation of the linear term	15
7.2	A uniform bound for the nonlinear remainder	16
8	Inverse in the multiplicity cap	17
9	Numbers and reproducible checks	18
10	Literature and the remaining questions	19
II	Fixed Taylor sectors for multiplicity cap growth constants	20
11	Objects and relationship to the main report	20
12	Exact positive sectors and the sharp remainder	21
13	Every algebraic order within a fixed sector	22
13.1	Separate the derivative correction	23
13.2	Uniform tail expansion and saddle localization	23
13.3	A finite algorithm for all coefficients	24
14	Reciprocation and the limits of exponential resolution	25
15	Independent checks and reproducibility	26
16	Questions left open	27

Guide to this report

This guide was written when the two manuscripts were merged. Everything after it, from Part I on, is the manuscripts' text, except for the passages marked **[write, 2 October 2026]** and the changes listed under “Conventions of the merge” below.

What the report proves

- **Part I** (source 47). Lemma 2.1 and Section 2: an exact compaction by a suffix involution, for every cap. Theorem 1.1: for every fixed $b \geq 3$ the exponential generating function has radius T_b , and $\log a_n^{(b)} = \log n! - n \log T_b + O_b(n/\log n)$, by uniform positive-operator barriers (Sections 3–4) and a coefficient extraction at every length (Section 5). Section 6: the length inverse $N_b(X) = n_0 + O_b(L/(\log L)^3)$, $n_0 = L/W_0(L/(eT_b))$. Theorem 1.2 and Section 7: monotone convergence $T_b \downarrow \pi^2/6$, the exact first variation $(b+1)(\zeta(b+2) - 1)$ and the explicit bound $0 \leq T_b - \pi^2/6 - (b+1)(\zeta(b+2) - 1) \leq 160b^{5/2}(4/9)^b$ for every $b \geq 3$. Section 8: the W_{-1} cap inverse with integer brackets (35), (36), and counterexamples to rounding by $\lceil b_0 \rceil$.
- **Part II** (source 52). Theorem 12.1: $T_b - \pi^2/6 = \sum_{r \geq 1} A_{r,b}$ with strictly positive sectors, for every $b \geq 2$; for fixed K the remainder $\mathcal{R}_{K,b}$ is positive and $\mathcal{R}_{K,b} \sim A_{K+1,b}$. Theorem 13.1: $A_{r,b} = C_r b^{(3-r)/2} \beta_r^b (\sum_j c_{r,j} b^{-j} + \dots)$ to every fixed order, with $\beta_r = (r/(r+1))^r$, rational

$c_{r,j}$, explicit $c_{r,1}$, three corrections for $r = 2, 3$, and $A_{1,b} = (b+1)(\zeta(b+2) - 1)$ exactly. Corollary 14.1: the same for $6/\pi^2 - \mu_b$.

Sources and provenance

The report was built from two manuscripts, both dated 1 October 2026, both delivered in the arrival commit 096ee7b87 of 2 October 2026 as batch 77, manuscripts 47 and 52, and placed in this directory by commit aa7345800 (batch 77P2). Neither pins a ProveIt commit. Source 47 identifies the cap-two report (batch-77 manuscript 37) by the SHA-256 of its delivered T_EX and PDF; source 52 identifies source 47 the same way. All four hashes match the delivered files of 096ee7b87 (checked at the merge). Author lines: “A proof and reproducible research report” (47) and “A reproducible large cap addendum” (52); no tool or person is named.

Part	Source	Archive and main file	Pages	Labels	Prefix
I	47	oeis-bounded-multiplicity-family-report.zip, bounded-multiplicity-report.tex (612 lines)	13	48	amc:
II	52	oeis-fixed-sector-addendum.zip, fixed-sector-addendum.tex (312 lines)	7	29	amc:sec:

Nesting. Source 52 is an addendum of source 47: it refines the large-cap analysis of T_b and takes the identification $\mu_b = 1/T_b$ and both inverses from source 47 (bibliography key `main`), without repeating their proofs. It does not bundle source 47; its archive carries four replay outputs byte-identical to its own saved results, which were not staged.

Where the merge had to choose. (i) The Parts follow the dependency order 47, 52. (ii) Both sources are printed in full; nothing of either is omitted except the title blocks and the bibliographies, which are merged below. (iii) Source 52 re-derives source 47’s first variation $(b+1)(\zeta(b+2) - 1)$ by a different computation; it is kept as a marked second proof. (iv) Source 47’s remainder bound $O(b^{5/2}(4/9)^b)$ is not sharp in its power of b ; source 52’s sharp asymptotic is printed beside it in dated notes, and both are kept, because source 47’s bound has explicit constants and its integer brackets use them. (v) No symbol was renamed; the notation table below separates the letters the two sources use differently.

How the Parts depend on one another

Part II is an analysis of the integral T_b alone; its only combinatorial input is Part I’s identification $\mu_b = 1/T_b$ (Theorem 1.1), used in Corollary 14.1. Part I’s Theorem 1.2 is the case $K = 1$ of Theorem 12.1 with the first sector (50), except that Part I also proves explicit constants, valid for every $b \geq 3$, and Part II does not. Neither Part strengthens the other’s results in the length n , and Part II proves no new inverse.

Relation to other reports

[**write, 2 October 2026**] *Sibling report on cap two.* The cap-two case $b = 2$ (OEIS A202058, ascent sequences avoiding 000) is the subject of the report `SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/a202058-ascent-000-growth`, opened by batch 77 from four manuscripts. Its Part I is the “separately audited cap-two report” that source 47 cites: its main theorem (label `a58:fd:thm:main` there) gives $T_2 = 3\pi^2/8$, the root limit and the error $O(n^{9/13}(\log n)^3)$, and its later Parts sharpen the cap-two asymptotics and prove the cap-two ratio limit. This report uses from it only T_2 and the case $b = 2$ of (2); source 47 proves nothing for $b = 2$ and its barrier proof excludes $b = 2$ (after (19)).

By the coordinator's decision at the merge the two are separate reports, not Parts of one: the sibling is about the fine asymptotics of a single cap, this report about the growth constants of all caps $b \geq 3$ and their dependence on b .

No other repository report concerns ascent sequences with a multiplicity cap. The 201-avoiding ascent sequences (OEIS A202062) and the 120-avoiding ones (A202061) have their own batch-77 reports in the same directory, `a202062-ascent-201-enumeration` and `a202061-ascent-120-deficit`; both cite the paper of Conway, Conway, Elvey Price and Guttmann [1], but they share no theorem with this report.

Notation

Each Part keeps its source's letters; no symbol was renamed. The two sources agree on E_b , T_b , μ_b , f , $\tau = \pi^2/6$ and $R_b(v) = e^v - E_b(v)$. Symbols not in the table are local to the Part that uses them. The table prints the tempting false reading beside the true one.

Symbol	Meaning and where	Do not read it as
L_b , $A_{1,b}$	$(b+1)(\zeta(b+2) - 1)$: Part I's first variation L_b (30) is Part II's first sector $A_{1,b}$ (50).	L is also the log-target $\log X$ in Section 6.
H_b , H_r	Part I: $H_b = T_b - \tau - L_b$, the nonlinear remainder, which is Part II's $\mathcal{R}_{1,b}$. Part II: $H_r = \sum_{j \leq r} 1/j$, the harmonic number.	H_3 of Part II is $11/6$, not a remainder. Part I also writes $H = h(i)$ and $H_q(v)$ in Section 5.
β	Part I, Section 2: $\beta_0, \dots, \beta_{m-1}$ are the remaining label budgets of a state. Part II: $\beta_r = (r/(r+1))^r$, the exponential base of sector r ($\beta_2 = 4/9$, $\beta_3 = 27/64$).	Not related.
m	Part I: the number of active labels of a state (Sections 2–5); in Section 7, $m = b+1$. Part II: $m = b+1$ throughout.	
q , p	Part I: $q = \log E_b(v)$ and $p_j = \log E_{b-j}(v)$, the characteristic curve; $p = p_1$. Part II: $q = b^{-1}$ and $p = (3-r)/2$ in the coefficient algorithm.	
c , C	Part I: generic constants, c_j , c_b in (9), $C(q)$ in (17). Part II: C_r and the rational coefficients $c_{r,j}$ of (46).	
D , d	Part I: the weight sum D of a state; $D = 9/\pi^4$ in (33); d_r , d_j , d , $d_b = \mu_\infty - \mu_b$. Part II: $D = 1 - t + tx$, D_r , $d_{r,j}$, the integers $d_{k,n}$, and a cutoff $d < 1$.	
J , K	Part I: a random index J and its centre $K = NM(q)$; K also a constant in (11). Part II: $J_{r,b}$, the derivative-free sector integral (52), and K , the number of sectors.	
a	Part I: the counts $a_n^{(b)}$. Part II: the counts too, and $a_r(x)$, the sector integrand (38), and a cutoff $a < s_r$.	
$\mathcal{R}$, $\mathcal{E}$	Part II: $\mathcal{R}_{K,b}$, the remainder after K sectors; $\tilde{\mathcal{R}}_{K,b}$ for μ_b . Part I: $\mathcal{E}(q)$, the barrier error (18).	
B	Part I: a constant in Section 5. Part II: Bernoulli numbers B_{2k} in the coefficient algorithm.	
ϕ , h , λ	Part I: $\phi(y)$ and $\phi(x) = x^{3/2}(8/9)^x$; $h(y)$ and $h(x) = D(x+1)2^{-x}$; $\lambda = \dot{q}D$. Part II: the phase $\phi_r(s)$ and $\lambda_r = -\phi_r''(s_r)$.	

Status, scope and non-claims

Neither manuscript states how it was produced; both reached the repository through its incoming channel for AI-assisted research deliveries and are treated as AI-assisted. Both are unrefereed: source 47 calls its audits “mathematical checks of this argument, not journal peer review”, and source 52’s verification records are research reviews, not formal verification. Their quadratures are high-precision numerical checks, not interval-certified enclosures. Nothing in this report is formalized in Lean or Rocq, and placing it in this repository confers no formal status.

The union of the sources’ non-claims, all kept in the Parts, is as follows.

- No fixed-cap prefactor, ratio limit or all-orders coefficient expansion in n for $b \geq 3$; the error $O_b(n/\log n)$ is deliberately coarse and is not the true subexponential scale; $a_n^{(b)} \sim C_b n! \mu_b^n$ is not implied.
- The limits in length and cap are kept separate: the large-cap expansion concerns the growth constant after the length limit; no uniform asymptotics for $b = b(n)$ and no exchange of limits.
- The length inverse is a controlled leading inverse, not an exact rounding rule. The cap inverse is an inverse for the cap of the limiting growth constant, not a cap-selection guarantee for finite-length counts; near integer transitions no unconditional rounding is justified.
- The sector results hold for a fixed number of sectors and a fixed algebraic order. No optimal truncation, summability, uniformity in $r = r(b)$, justified sum of all sector expansions, or theorem at the accumulation scale e^{-b} ; finite truncations of an earlier sector do not resolve later sectors; the reciprocal remainder is positive only eventually for each fixed K ; no new inverse. The word “transseries” in Part II means only the exact positive-sector decomposition.
- Priority is not established; the literature checks are scoped. The cap-two compacted recurrence and the constant $8/(3\pi^2)$ (conjectured) are due to Conway, Conway, Elvey Price and Guttmann; source 47’s involution generalizes their compaction mechanism. The standalone OEIS page of A317784 could not be retrieved by source 47; its identification rests on A294220’s cross-reference.

Instances of results elsewhere in the repository

Several steps are instances of results proved in the transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/`. No novelty is claimed for these mechanics.

- The length inverse $n_0 = L/W_0(L/(eT_b))$ solves $x \log(x/(eT_b)) = L$; with $Y = \log(x/(eT_b))$ this is $Y + \log Y = \log(L/(eT_b))$, the linear–logarithmic equation $aX + b \log X = L'$ with $a = b = 1$, solved on the principal branch by the volume’s Lambert core, theorem `p0:thm:lambert-core`.
- The cap model $D(b+1)2^{-b} = \varepsilon$ is, in $X = b+1$, the equation $(\log 2)X - \log X = \log(2D/\varepsilon)$, the case $a > 0 > b$ of the same theorem, whose branch rule selects W_{-1} for the large root, as (34) does.
- The integer brackets (35) and (36) are instances of the separation step of the volume’s theorem `p0:thm:staircase`. That step’s separation inequality is formalized in Lean as `staircase_separation`, in the file `StaircaseInversion.lean` of the directory `Analysis/FabiusFunction/Lean/FabiusFunction/`. Nothing specific to ascent sequences or to T_b is formalized.

Conventions of the merge

- Every label carries the report prefix: `amc:` in Part I and `amc:sec:` (“sectors”) in Part II. Source 47’s own section labels began with `sec:`, so its section labels read `amc:sec:compaction`, `amc:sec:checks` and so on; they belong to Part I. Part II’s labels all have the form `amc:sec:eq:...`, `amc:sec:thm:...`, `amc:sec:cor:...` or `amc:sec:sec:...`. Guide labels are `amc:guide:`, Part labels `amc:part:`.
- Sections are numbered through the report: source 47’s Section k is Section k here (Sections 1–10), source 52’s is Section $k + 10$ (Sections 11–16). Theorems are numbered within sections, as in both sources, so source 52’s Theorem 2.1 is Theorem 12.1. Equations are numbered through the report.
- Source 52’s citation `\cite{main}` and its phrase “the separately supplied report” are now references to Part I; elsewhere “the main report” means Part I, with references added where it names a result. Its bibliography entry `main` is dropped. Source 52’s abstract is printed as delivered except for a bracketed pointer to Part I.
- Sentences about “the accompanying source archive”, “the package”, “the ZIP”, `verify.sh`, `build.sh`, `checks/`, `sources/` or `provenance.json` describe the delivered archives, which survive in commit `096ee7b87`. Which of their files this report ships, and under which names, is listed in the README; bracketed notes in the bibliography give the shipped names of the files it cites.

Part I

Factorial growth of ascent sequences with bounded multiplicity

Source. Batch-77 manuscript 47, archive `oeis-bounded-multiplicity-family-report.zip`, main file `bounded-multiplicity-report.tex` (612 lines, 13-page PDF), dated 1 October 2026. Author line: “A proof and reproducible research report”. No commit pin; it identifies the cap-two report by file hashes. Printed in full, with labels prefixed `amc::`; one section label and the `[write]` notes were added at the merge.

Abstract (as delivered). Let $a_n^{(b)}$ count ascent sequences of length n in which each value occurs at most b times. For every fixed $b \geq 3$ we prove the exact factorial root constant

$$\lim_{n \rightarrow \infty} \left(\frac{a_n^{(b)}}{n!} \right)^{1/n} = \left[\int_0^\infty \frac{\log E_b(v)}{E_b(v) - 1} dv \right]^{-1}, \quad E_b(v) = \sum_{j=0}^b \frac{v^j}{j!}.$$

The separate cap-two proof gives the same formula. We establish exact compaction by a suffix involution, uniform positive-operator barriers, and a coefficient argument for every n . A non-optimal logarithmic error supports a controlled inverse for the required length. We also prove a large-cap expansion of the constant, including an exact zeta first variation and an exponentially smaller remainder, and obtain a rounding-safe cap inverse. This report does not determine fixed-cap prefactors, ratio limits, or all-orders coefficient expansions. The limits in length and cap are kept separate.

1 Results and interpretation

An ascent sequence is a word $w_1 \cdots w_n$ of nonnegative integers with $w_1 = 0$ and

$$0 \leq w_i \leq 1 + \text{asc}(w_1 \cdots w_{i-1}) \quad (i \geq 2), \quad \text{asc}(w) = \#\{i : w_i < w_{i+1}\}.$$

Write $a_0^{(b)} = 1$ and let $a_n^{(b)}$ count the words with every label used at most b times. Equivalently, these words avoid the constant pattern of length $b+1$. This is the array in OEIS A294220; cap two is A202058 and cap three is A317784 [3, 4].

Theorem 1.1. *For every fixed integer $b \geq 3$, set*

$$T_b = \int_0^\infty \frac{\log E_b(v)}{E_b(v) - 1} dv, \quad \mu_b = T_b^{-1}. \quad (1)$$

The integrand is interpreted as 1 at zero. The exponential generating function $A_b(t) = \sum_{n \geq 0} a_n^{(b)} t^n / n!$ has radius T_b , and

$$\log a_n^{(b)} = \log(n!) - n \log T_b + O_b\left(\frac{n}{\log n}\right). \quad (2)$$

Consequently $(a_n^{(b)} / n!)^{1/n} \rightarrow \mu_b$ and

$$a_n^{(b)} = n! \mu_b^n \exp\left(O_b\left(\frac{n}{\log n}\right)\right).$$

The subscript b permits constants and starting indices to depend on the fixed cap. The logarithmic error is deliberately coarse; it is not an estimate of the true subexponential scale. In particular, the theorem does not imply $a_n^{(b)} \sim C_b n! \mu_b^n$ or $a_{n+1}^{(b)} / ((n+1)a_n^{(b)}) \rightarrow \mu_b$.

The separately audited cap-two report [5] proves $T_2 = 3\pi^2/8$ and a stronger logarithmic error $O(n^{9/13}(\log n)^3)$, so (2) also holds for $b = 2$. Its analytic proof is not repeated here. For $b = 1$, the unique word at each length is strictly increasing, $T_1 = \infty$, and the normalized root is zero.

[write, 2 October 2026] *The cap-two report in this repository.* The report cited here is batch-77 manuscript 37, printed as Part I of the sibling report `a202058-ascent-000-growth` in this collection directory; its main theorem (label `a58:fd:thm:main` there) proves that the exponential generating function of A202058 has radius $T = 3\pi^2/8$, the root limit $(a_n/n!)^{1/n} \rightarrow 8/(3\pi^2)$, and the error $O(n^{9/13}(\log n)^3)$. The SHA-256 values that source 47 records for that report's delivered `a202058-report.tex` and `.pdf` (`data/47-caps-provenance.json`) match the files of the arrival commit `096ee7b87`; this was checked at the merge. Later Parts of the sibling report sharpen the cap-two error and prove the cap-two ratio limit (see its README). The two reports are kept separate, by the coordinator's decision at the merge: the sibling treats the fine asymptotics of the single cap $b = 2$, this one the growth constants of every cap $b \geq 3$ and their large-cap behaviour, and this report takes from the sibling only T_2 and the $b = 2$ case of (2).

Theorem 1.2. *As $b \rightarrow \infty$ through integers,*

$$T_b = \frac{\pi^2}{6} + (b+1)(\zeta(b+2) - 1) + O\left(b^{5/2}(4/9)^b\right), \quad (3)$$

$$T_b - \frac{\pi^2}{6} \sim \frac{b+1}{2^{b+2}}, \quad \frac{6}{\pi^2} - \mu_b \sim \frac{9(b+1)}{\pi^4 2^b}. \quad (4)$$

Moreover T_b strictly decreases to $\pi^2/6$, and μ_b strictly increases to $6/\pi^2$.

This is an expansion of the fixed-cap growth constant after the length limit. It does not assert uniform coefficient asymptotics when $b = b(n)$ grows. Sections 6 and 8 give different inverses for the length and the cap.

[write, 2 October 2026] Part II (source 52) sharpens the remainder in (3). The exact identity $T_b - \pi^2/6 = \sum_{r \geq 1} A_{r,b}$ (Theorem 12.1) has first sector $A_{1,b} = (b+1)(\zeta(b+2) - 1)$, the first variation of (3), and its remainder after one sector is asymptotic to the second sector:

$$T_b - \frac{\pi^2}{6} - (b+1)(\zeta(b+2) - 1) \sim A_{2,b} \sim \frac{8}{9\sqrt{\pi}} \sqrt{b} \left(\frac{4}{9}\right)^b$$

(equations (43) and (48)). So the exponential rate $(4/9)^b$ of (3) is sharp, while its power $b^{5/2}$ is not: the true power is $b^{1/2}$. The two statements are consistent. The bound of this Part has explicit constants and holds for every $b \geq 3$ (Section 8); Part II's is sharp but asymptotic, with constants that are not explicit.

2 Exact compaction and the counting operator

For any ascent-sequence prefix, $\max(w) \leq \text{asc}(w)$. Indeed a new maximum creates an ascent, whereas a nonascent cannot raise the maximum. Thus the top admissible label $1 + \text{asc}(w)$ is unused and strictly larger than the last value.

Delete labels whose budgets have been exhausted. The remaining active labels, in numerical order, have positive budgets $\beta_0, \dots, \beta_{m-1} \in \{1, \dots, b\}$. Let k be the number of active labels at or below the actual last value. This can be a virtual boundary when that value has just been deleted. Choosing active rank i is an ascent exactly when $i \geq k$. Decrease β_i by one; remove it and set $k' = i$ if it reaches zero, and otherwise set $k' = i + 1$. An ascent creates precisely one additional top label of budget b .

Lemma 2.1 (Adjacent-budget interchange). *Suppose $x < y$ are adjacent active labels with both at or below the last value. Exchanging their positive remaining budgets, while retaining the last boundary, preserves the number of legal suffixes of every length.*

Proof. In a suffix, reverse each maximal contiguous block consisting only of x, y , and then interchange x and y throughout that block. This is an involution that exchanges their total usages. Every intermediate usage is at most the total, so the exchanged quotas hold at every stage.

Inside a binary block, a comparison $p < q$ becomes the same comparison between the complemented pair in reverse order. Its total number of ascents is preserved. No possible outside label lies strictly between x and y : the two labels were adjacent among the initially active labels, exhausted labels never return, and future labels are all above the initial alphabet. Thus boundary comparisons with outside letters are unchanged. Entry into an initial binary block is nonascending both before and after the map, since the initial last value is at least y .

Ascent totals therefore agree at each block end and before every unchanged outside letter. The timing inside a block can change, but both block labels were already admissible initially, and the admissible upper bound never decreases. This proves legality as well as invertibility. $\square$

The lemma allows permutations below the boundary, not arbitrary sorting above it. Start with a canonical state in which budgets increase with rank. After choosing budget $r > 1$, move its decremented entry left to the beginning of the old budget- r group. Every swap lies among the first $i + 1$ entries, hence below the new boundary $i + 1$, which stays fixed. Exhaustion leaves an already sorted tuple and boundary i . A new top label is appended. This gives an exact reduction at every step. The last boundary does not follow the moved budget.

Write $N_0 = u$ for the number of unused labels and $N_j = s_j$ for those used j times, $1 \leq j < b$. Groups occur in increasing rank order as

$$N_{b-1}, N_{b-2}, \dots, N_1, N_0, \quad m = \sum_{j=0}^{b-1} N_j, \quad u \geq 1, \quad 0 \leq k < m.$$

If rank i belongs to group j , put $\alpha = \mathbf{1}_{i \geq k}$. Its child is

$$N' = N - e_j + \mathbf{1}_{j+1 < b} e_{j+1} + \alpha e_0, \quad k' = \begin{cases} i, & j = b - 1, \\ i + 1, & j < b - 1. \end{cases} \quad (5)$$

These transitions preserve $u \geq 1$ and $k < m$: a choice of the final unused top label is always ascending and creates another. They satisfy $m' \leq m + 1$. Define the positive operator and its series by

$$(\mathcal{T}f)(x) = \sum_{i=0}^{m-1} f(x_i), \quad F(t, x) = \sum_{r \geq 0} \mathcal{T}^r \mathbf{1}(x) \frac{t^r}{r!}. \quad (6)$$

Let x_1 have $s_1 = u = k = 1$ and all other $s_j = 0$. Then

$$\mathcal{T}^r \mathbf{1}(x_1) = a_{r+1}^{(b)}, \quad F(t, x_1) = A'_b(t).$$

All series identities below allow $+\infty$ and use nonnegative sums when necessary.

3 Characteristic curve and a uniform barrier

For this section fix $b \geq 3$ and write $\theta = (b-1)/b > 1/2$. Parameterize time by $v \geq 0$:

$$q = \log E_b(v), \quad p_j = \log E_{b-j}(v) \quad (1 \leq j < b), \quad p_0 = q, \quad p_b = 0, \quad t(v) = \int_0^v \frac{\log E_b(z)}{E_b(z) - 1} dz. \quad (7)$$

Its endpoint is $T_b < \infty$: the integrand is $O_b(\log v/v^b)$ at infinity. A dot denotes a time derivative. Direct differentiation gives

$$\dot{q} = e^{p_1} \frac{1 - e^{-q}}{q}, \quad \dot{p}_j = \frac{e^q - 1}{q} e^{p_{j+1} - p_j}. \quad (8)$$

At zero the removable values are all 1, so every p_j, q is strictly increasing. Polynomial asymptotics give

$$p_j = \frac{b-j}{b} q + c_j + O_b(e^{-q/b}), \quad (9)$$

$$e^{p_1} \sim c_b e^{\theta q}, \quad \dot{q} \sim c_b e^{\theta q}/q, \quad c_b = \frac{(b!)^\theta}{(b-1)!}. \quad (10)$$

Fixed-order differentiated estimates follow from the underlying rational functions of v , rather than differentiation of an arbitrary error term.

Define

$$\rho_0 = 1, \quad \rho_b = 0, \quad \rho_j = \frac{\dot{p}_j}{\dot{q}} = \frac{E_b(v)E_{b-j-1}(v)}{E_{b-1}(v)E_{b-j}(v)} \quad (1 \leq j < b).$$

They satisfy $1 = \rho_0 \geq \rho_1 \geq \dots \geq \rho_{b-1} > 0$. To see this, put $d_r = v^r/r!$ and note

$$E_r^2 - E_{r-1}E_{r+1} = d_r \left(E_r - \frac{v}{r+1} E_{r-1} \right) \geq 0;$$

the bracket has positive coefficients. Also $\rho_j \rightarrow (b-j)/b$. Positivity, continuity, and rational asymptotics yield fixed-cap constants $c > 0, K < \infty$ such that

$$c \leq \rho_j \leq 1, \quad |(\rho_j)_q| \leq K \quad (0 \leq j < b, q \geq 0). \quad (11)$$

For a state x , let

$$D = \sum_{j=0}^{b-1} \rho_j N_j, \quad h(y) = \int_0^y \rho_{\text{group}(z)} dz, \quad r(y) = h(y)/D, \quad r = r(k),$$

where group weights are constant on unit rank intervals. Put

$$\psi(t, x) = \exp \left(\sum_{j=1}^{b-1} p_j s_j + qu - qr \right). \quad (12)$$

Here $D \geq u \geq 1$, $cm \leq D \leq m$, $0 \leq r \leq 1$, and $\psi(0, x) = 1$.

3.1 Exact child rank and its bounds

If rank i belongs to group j , write $H = h(i)$ and $d_j = \rho_j - \rho_{j+1} \in [0, 1]$. Every compaction swap is below the new boundary, so its weight sum is unchanged. Consequently

$$D_i = D - d_j + \alpha, \quad h_{x_i}(k_i) = H + \rho_{j+1}, \quad r_i = \frac{H + \rho_{j+1}}{D - d_j + \alpha}. \quad (13)$$

This includes saturation through $\rho_b = 0$. Since $D_i \geq 1$ and $D_i \geq D - 1$, we have $D_i \geq D/2$ and

$$|r_i - r(i)| \leq \frac{4}{D}. \quad (14)$$

For an ascent, $i \geq k$ and $D_i \leq D + 1$, whence the crucial one-sided bound

$$r(k) - r_i \leq \frac{H}{D} - \frac{H}{D+1} \leq \frac{1}{D+1} \leq \frac{1}{2}. \quad (15)$$

3.2 Frozen integral and residual

For real y in group j set

$$g(y) = \exp(p_{j+1} - p_j + q\mathbf{1}_{y \geq k} - q(r(y) - r(k))).$$

The exact transition ratio $\psi(t, x_i)/\psi(t, x)$ replaces $r(i)$ by r_i in $g(i)$. The characteristic equations imply

$$\int_0^m g(y) dy = \dot{q}D = \sum_{j=1}^{b-1} \dot{p}_j s_j + \dot{q}u =: \lambda. \quad (16)$$

Indeed, for $\phi(y) = e^{-qh(y)/D}$ one has $\phi_y = -(q\rho_j/D)\phi$ and $e^{p_{j+1}-p_j} = q\dot{q}\rho_j/(e^q - 1)$. Using $\phi(0) = 1, \phi(m) = e^{-q}$, the integrals before division by $\phi(k)$ are

$$\frac{\dot{q}D}{e^q - 1}(1 - \phi(k)), \quad \frac{\dot{q}De^q}{e^q - 1}(\phi(k) - e^{-q}),$$

whose sum is $\dot{q}D\phi(k)$. Continuity handles $q = 0$.

By (9), every frozen ratio is at most $C_b e^{\theta q}$. On each of at most $b + 1$ intervals cut at group boundaries and k, g decreases. All endpoints are integers; hence the left-sum error is bounded by the sum of left endpoint values:

$$\left| \sum_i g(i) - \lambda \right| \leq C_b e^{\theta q}.$$

Exact descending ratios obey the same bound; exact ascending ratios are at most $C_b e^{(\theta+1/2)q}$ by (15). Apply $|e^z - e^w| \leq |z - w| \max(e^z, e^w)$, (14), and $m/D \leq 1/c$ to obtain

$$\left| \frac{\mathcal{T}\psi}{\psi} - \sum_i g(i) \right| \leq C_b q e^{(\theta+1/2)q}.$$

Finally $|h_q|, |D_q| \leq Km$ imply $|r_q| \leq 2K/c$. Thus

$$\frac{\partial_t \psi}{\psi} = \lambda - \dot{q}(r + qr_q), \quad \left| \frac{\partial_t \psi}{\psi} - \lambda \right| \leq C_b(1 + q)e^{\theta q}.$$

Together these give the state-uniform estimate

$$\left| \frac{\partial_t \psi}{\psi} - \frac{\mathcal{T} \psi}{\psi} \right| \leq C_b(1+q)e^{(\theta+1/2)q} =: C(q). \quad (17)$$

Let

$$\mathcal{E}(q) = \int_0^q \frac{C(z)}{\dot{q}(z)} dz = O_b((1+q)^3 e^{q/2}), \quad g_{\pm} = e^{\pm \mathcal{E}} \psi. \quad (18)$$

Then $\partial_t g_- \leq \mathcal{T} g_-$ and $\partial_t g_+ \geq \mathcal{T} g_+$, and both start at 1. Since $\theta > 1/2$,

$$\mathcal{E}(q) = o(e^{p_1(q)}). \quad (19)$$

This strict inequality explains why this particular proof excludes $b = 2$.

4 Comparison on the infinite state space

Let X be the continuous-time chain with each rank-choice edge at rate 1. Its generator is $L = \mathcal{T} - mI$ and its total rate is m . Because m grows by at most one per jump, comparison with a linear pure-birth process proves nonexplosion. Expanding by finite jump paths cancels survival factors against the potential factor and gives the extended-real identity

$$F(t, x) = \mathbb{E}_x \exp \left(\int_0^t m(X_s) ds \right). \quad (20)$$

Let τ_M be the first exit from $m \leq M$ and $Y_s = \exp(\int_0^s m(X_r) dr)$. The stopped region is finite. Applying stopped Dynkin comparison to $Y_s g_+(t-s, X_s)$ and discarding its nonnegative exit term gives $F(t, x) \leq g_+(t, x)$ upon letting $M \rightarrow \infty$.

For the lower bound fix $\delta > 0$ with $t + \delta < T_b$. Strict increase of all p_j, q gives an $\eta > 0$, uniform for $0 \leq v \leq t$, such that

$$g_-(v, y) \leq C e^{-\eta m(y)} g_+(v + \delta, y). \quad (21)$$

Indeed the population terms each decrease by at least η in the log ratio; its rank term is bounded above by $q(t + \delta)$, and the $\mathcal{E}$ terms have favorable signs. The stopped subsolution inequality is

$$g_-(t, x) \leq \mathbb{E}_x[Y_t; \tau_M > t] + \mathbb{E}_x[Y_{\tau_M} g_-(t - \tau_M, X_{\tau_M}); \tau_M \leq t].$$

On exit $m(X_{\tau_M}) = M + 1$. Equation (21) and the stopped supersolution inequality at horizon $t + \delta$ bound the second term by $C e^{-\eta(M+1)} g_+(t + \delta, x) \rightarrow 0$. Nonexplosion handles the first term. We have proved

$$e^{-\mathcal{E}(q)} \psi(t, x) \leq F(t, x) \leq e^{\mathcal{E}(q)} \psi(t, x), \quad 0 \leq t < T_b. \quad (22)$$

No uniqueness assertion for an infinite differential system is required.

5 The radius and the coefficient limit

Let x_N have $s_1 = N, u = 1, k = N$ and all other populations zero; the increasing prefix of length N reaches it. Its rank formula yields

$$\log \psi(t, x_N) = N p_1(q) + \frac{q}{\rho_1 N + 1}, \quad \log F(t, x_N) = N p_1(q) + O_b(\mathcal{E}(q) + q), \quad (23)$$

with error independent of N . In particular $F(t, x_N) \geq e^{Np_1 - \mathcal{E}}$. The positive semigroup identity and one increasing-prefix path at every length give, for $h > 0$,

$$\begin{aligned} F(t+h, x_1) &= \sum_{r \geq 0} \frac{h^r}{r!} (\mathcal{T}^r F(t, \cdot))(x_1) \\ &\geq \sum_{r \geq 0} \frac{h^r}{r!} F(t, x_{r+1}) \geq \exp(p_1 - \mathcal{E} + he^{p_1}). \end{aligned}$$

As $t \uparrow T_b$, (19) makes the last expression diverge. Monotonicity forces $F(T_b + h, x_1) = \infty$. Equation (22) gives finiteness below T_b . Thus $F(t, x_1)$ and A_b have exact radius T_b , proving the normalized-root limsup. The next argument is needed for the full limit.

5.1 A concentration estimate from barrier values

Put $p = p_1$, $d = (\log t)_q$ and $M(q) = p_q/d = tp$. Direct characteristic asymptotics give

$$p_q \rightarrow \theta, \quad p_{qq} \rightarrow 0, \quad d \sim \frac{qe^{-\theta q}}{c_b T_b}, \quad d'/d \rightarrow -\theta, \quad M(q) \sim \frac{\theta c_b T_b e^{\theta q}}{q}. \quad (24)$$

These limits are uniform after bounded shifts in q . For $c_j = \mathcal{T}^j \mathbf{1}(x_N)$ define

$$\mathbb{P}(J = j) = \frac{c_j t(q)^j}{j! F(t(q), x_N)}, \quad K = NM(q).$$

The deterministic center K need not be the exact mean. We use values of (23), never derivatives of its error. Set

$$H_q(v) = p(q+v) - p(q) - M(q)(\log t(q+v) - \log t(q)).$$

For $|v| \leq 1$, $H_q(0) = H'_q(0) = 0$ and

$$H''_q(v) \rightarrow \theta^2 e^{-\theta v}, \quad M(q)d(q+v) \rightarrow \theta e^{-\theta v}$$

uniformly. Consequently there are fixed-cap constants $B, c > 0$ with

$$|H_q(v)| \leq Bv^2, \quad M(q)|\log t(q+v) - \log t(q)| \geq c|v|.$$

The exponential Markov bounds at $q \pm \gamma\varepsilon$, with fixed small $\gamma > 0$, now give

$$\mathbb{P}(|J - K| > \varepsilon K) \leq 2 \exp\{-c'_b N \varepsilon^2 + 2\mathcal{E}(q+1) + O_b(q+1)\} \quad (25)$$

for sufficiently large q and $0 < \varepsilon < 1/2$. Explicitly the main exponent is at most $N(B\gamma^2\varepsilon^2 - c\gamma\varepsilon^2)$; choosing γ small makes it negative. Uniformity permits shrinking ε .

5.2 Extraction for each sufficiently large length

For every sufficiently large integer n , choose

$$q = \frac{1}{2} \log n, \quad \varepsilon = (\log n)^{-2}, \quad N = \left\lfloor \frac{(1 - 4\varepsilon)n}{M(q)} \right\rfloor.$$

Then

$$\begin{aligned} N &\asymp_b n^{1-\theta/2} \log n, & N\varepsilon^2 &\asymp_b \frac{n^{1-\theta/2}}{(\log n)^3}, \\ K &= (1-4\varepsilon)n + O_b(n^{\theta/2}/\log n), & \mathcal{E}(q+1) &= O_b(n^{1/4}(\log n)^3). \end{aligned}$$

Since $1/2 < \theta < 1$, $N\varepsilon^2$ dominates the error and both N and the floor loss in K are $o(\varepsilon n)$. Equation (25) puts probability at least $1/2$ in $|J - K| \leq \varepsilon K$. Some integer j there satisfies

$$\frac{c_j t(q)^j}{j!} \geq \frac{F(t(q), x_N)}{4\varepsilon K + 6}.$$

For large n it obeys $(1-6\varepsilon)n \leq j$ and $N+j \leq n$, giving

$$\log c_j \geq \log(j!) - j \log t(q) + Np(q) - \mathcal{E}(q) - O_b(\log n). \quad (26)$$

The increasing prefix reaches x_N in $N-1$ transitions, so $a_{N+j}^{(b)} \geq c_j$. Appending the new maximum $1 + \text{asc}(w)$ repeatedly gives an injective extension to length n , so $a_n^{(b)} \geq c_j$.

5.3 A quantitative logarithmic error

From (8),

$$T_b - t(q) = O_b((1+q)e^{-\theta q}). \quad (27)$$

In (26) the term $Np(q)$ is nonnegative. Also $\log(n!/j!) \leq 6\varepsilon n \log n = 6n/\log n$ and

$$|-j \log t(q) + n \log T_b| = O_b(\varepsilon n + n(1+q)e^{-\theta q}).$$

All latter errors, and $\mathcal{E}(q)$, are $o_b(n/\log n)$ for $q = \frac{1}{2} \log n$. Hence

$$\log a_n^{(b)} \geq \log(n!) - n \log T_b - O_b(n/\log n).$$

For the matching upper estimate use a single coefficient of $F(t, x_1)$:

$$a_n^{(b)} \leq (n-1)! F(t(q), x_1) t(q)^{-(n-1)}.$$

Equations (23) and (27) bound its excess logarithm above $\log(n!) - n \log T_b$ by

$$O_b(\mathcal{E}(q) + q + \log n + n(1+q)e^{-\theta q}) = o_b(n/\log n).$$

This proves (2), and therefore the full root limit. The proof uses no unproved ratio asymptotic.

6 Inverse in the sequence length

For fixed $b \geq 2$ and $X > 1$, define the integer threshold

$$N_b(X) = \min\{n \geq 0 : a_n^{(b)} \geq X\}, \quad L = \log X.$$

The injective extension above makes the counts nondecreasing, and Theorem 1.1 makes them unbounded. Let W_0 denote the principal real branch of Lambert W , characterized by $W_0(z)e^{W_0(z)} = z$ for $z > 0$. As $X \rightarrow \infty$ with b fixed,

$$n_0 = \frac{L}{W_0(L/(eT_b))}, \quad N_b(X) = n_0 + O_b\left(\frac{L}{(\log L)^3}\right) = n_0 \left(1 + O_b((\log L)^{-2})\right). \quad (28)$$

To prove it, Stirling and (2) give $\log a_n^{(b)} = G(n) + O_b(n/\log n)$ for $G(x) = x \log(x/(eT_b))$. The threshold inequalities at $N = N_b(X)$ and $N - 1$, together with $G(N) - G(N - 1) = O_b(\log N)$, imply $L = G(N) + O_b(N/\log N)$. Thus $N \sim L/\log L$, while $G(n_0) = L$ exactly. Since $G'(x) = \log(x/T_b) \asymp \log L$ between comparable N and n_0 , the mean value theorem gives (28).

This is a controlled leading inverse, not an exact rounding rule: its error tends to infinity. One may expand W_0 itself in logarithms, but such a formal expansion cannot improve the error inherited from the unknown coefficient correction. For $b = 2$, the separate report gives a sharper inverse using its stronger logarithmic bound.

7 Large caps and an exact first variation

Put $f(x) = \log x/(x - 1)$, continuously extended at 1, and $\tau = \pi^2/6$. The representation

$$f(x) = \int_0^1 \frac{dt}{1 + t(x - 1)} \quad (29)$$

shows $f' < 0$ and $f'' > 0$ on $[1, \infty)$. Thus E_b increasing strictly in b implies T_b decreasing strictly. The cap-two integrand dominates all later integrands and is integrable. Dominated convergence and the nonnegative geometric series give

$$T_b \downarrow \int_0^\infty \frac{v}{e^v - 1} dv = \sum_{k \geq 1} \frac{1}{k^2} = \tau.$$

Write

$$R_b(v) = e^v - E_b(v), \quad \Delta_b(v) = f(E_b(v)) - f(e^v), \quad L_b = \int_0^\infty [-f'(e^v)] R_b(v) dv.$$

Convexity gives $\Delta_b(v) \geq [-f'(e^v)] R_b(v) \geq 0$. We will prove

$$L_b = (b + 1)(\zeta(b + 2) - 1), \quad 0 \leq T_b - \tau - L_b = O(b^{5/2}(4/9)^b). \quad (30)$$

7.1 Evaluation of the linear term

For $v > 0$,

$$-f'(e^v) = \sum_{h \geq 2} ((h - 1)v - 1)e^{-hv}, \quad R_b(v) = \sum_{k \geq b+1} \frac{v^k}{k!}.$$

Although individual terms can have either sign near zero, their integrated absolute values are bounded by

$$\sum_{k \geq b+1} \sum_{h \geq 2} \left(\frac{(h - 1)(k + 1)}{h^{k+2}} + \frac{1}{h^{k+1}} \right) < \infty \quad (b \geq 2).$$

Fubini is therefore legitimate. Integration and then summation in k yield

$$L_b = \sum_{h \geq 2} \sum_{k \geq b+1} \frac{(h - 1)k - 1}{h^{k+2}} = \sum_{h \geq 2} \frac{b + 1}{h^{b+2}} = (b + 1)(\zeta(b + 2) - 1).$$

7.2 A uniform bound for the nonlinear remainder

There is an absolute constant C with

$$0 \leq f''(z) \leq C \frac{1 + \log z}{z^3} \quad (z \geq 1). \quad (31)$$

This follows from smoothness at 1 and the derivative asymptotics at infinity, or directly from (29). For $0 \leq v \leq b$, the function $e^{-v}E_b(v)$ decreases, since its derivative is $-e^{-v}v^b/b!$. Stirling therefore gives

$$e^{-v}E_b(v) \geq e^{-b}E_b(b) \geq e^{-b}b^b/b! \geq cb^{-1/2}$$

with absolute $c > 0$. Taylor's theorem on $[E_b(v), e^v]$, using $\log z \leq v$, gives

$$0 \leq \Delta_b(v) - [-f'(e^v)]R_b(v) \leq Cb^{3/2}(1+v)e^{-3v}R_b(v)^2. \quad (32)$$

Let $m = b + 1$. Expanding the square, grouping $S = k + \ell$, and using $\sum_{k=m}^{S-m} \binom{S}{k} \leq 2^S$, we obtain

$$\begin{aligned} \int_0^\infty (1+v)e^{-3v}R_b(v)^2 dv &\leq \sum_{S \geq 2m} 2^S \left(\frac{1}{3^{S+1}} + \frac{S+1}{3^{S+2}} \right) \\ &= O\left(b(4/9)^b\right). \end{aligned}$$

All these integrands and terms are nonnegative, so Tonelli applies. This bounds the integrated remainder on $[0, b]$ by $O(b^{5/2}(4/9)^b)$.

For $v \geq b \geq 3$, $E_b(v) \geq v^b/b! \geq 2$ and $\log E_b(v) \leq v$. Convexity implies the remainder is at most $\Delta_b(v) \leq f(E_b(v))$, and hence its tail is at most

$$2b! \int_b^\infty v^{1-b} dv = \frac{2b!b^{2-b}}{b-2} = O(b^{3/2}e^{-b}) = O\left(b^{5/2}(4/9)^b\right).$$

This proves (30) and (3) with constants independent of b .

Since

$$L_b = \frac{b+1}{2^{b+2}} + O(b3^{-b}),$$

the remainder is smaller than the leading term by an exponentially decaying factor. The reciprocal identity

$$\mu_\infty - \mu_b = \frac{T_b - \tau}{\tau T_b}, \quad \mu_\infty = \frac{1}{\tau} = \frac{6}{\pi^2},$$

then gives the useful quantified form

$$\mu_\infty - \mu_b = D(b+1)2^{-b} \left[1 + O\left(b^{3/2}(8/9)^b\right) \right], \quad D = \frac{9}{\pi^4}. \quad (33)$$

Indeed the additive contributions $O(b3^{-b})$ and $O(b^24^{-b})$ are absorbed by $O(b^{5/2}(4/9)^b)$. This also proves (4). Equation (3) is a rigorously controlled first-variation expansion; it is not an all-orders transseries.

8 Inverse in the multiplicity cap

For small $\varepsilon > 0$ let

$$b_\varepsilon = \min\{b \geq 2 : \mu_b \geq \mu_\infty - \varepsilon\}.$$

Monotonicity proves existence. The large real solution of the leading model $D(b+1)2^{-b} = \varepsilon$ is

$$b_0 = -\frac{1}{\log 2} W_{-1}\left(-\frac{\varepsilon \log 2}{2D}\right) - 1, \quad (34)$$

where W_{-1} is the real branch with $W_{-1}(z) \leq -1$ on $[-1/e, 0)$. The branch is essential; W_0 gives the irrelevant small root.

There are constants C and $\varepsilon_0 > 0$ such that, for $0 < \varepsilon < \varepsilon_0$, with

$$\eta = C b_0^{3/2} (8/9)^{b_0},$$

one has the integer-safe bracket

$$\lceil b_0 - \eta \rceil \leq b_\varepsilon \leq \lceil b_0 + \eta \rceil. \quad (35)$$

No interpolation of the exact sequence T_b to real b is assumed.

Here is the discrete justification. For the real leading model $h(x) = D(x+1)2^{-x}$,

$$(\log h)'(x) = \frac{1}{x+1} - \log 2$$

is negative and bounded away from zero for large x . Equation (33) first places b_ε within a fixed bounded distance of b_0 : at integers several units below or above b_0 , the fixed change of h dominates the vanishing relative error. In that bounded window the relative error is uniformly $O(b_0^{3/2}(8/9)^{b_0})$. Increasing C if necessary, every integer $b < b_0 - \eta$ fails the threshold and every integer $b \geq b_0 + \eta$ meets it. Monotonicity handles all integers outside the window, giving (35).

A useful explicit strengthening, proved by the same estimates, is

$$\lceil b_0 \rceil + 1 \leq b_\varepsilon \leq \lceil b_0 + 2000 b_0^{3/2} (8/9)^{b_0} \rceil \quad (b_0 \geq 13). \quad (36)$$

Here are details to make its constants and strict lower bound reproducible. The bound $f''(z) \leq 16(1 + \log z)/z^3$ and $b! \leq 3\sqrt{b}(b/e)^b$ make (32) valid with 216. Its integrated central bound is $64b^{3/2}(b+4)(4/9)^b$, and its tail is at most $6b^{5/2}(4/9)^b$. Thus $0 \leq H_b := T_b - \tau - L_b \leq 160b^{5/2}(4/9)^b$ for every $b \geq 3$.

[write, 2 October 2026] In Part II's notation, $L_b = A_{1,b}$ and $H_b = \mathcal{R}_{1,b}$, and Theorem 12.1 with (48) gives $H_b \sim A_{2,b} \sim \frac{8}{9\sqrt{\pi}} \sqrt{b} (4/9)^b$. The explicit bound $160b^{5/2}(4/9)^b$ above is therefore valid but not sharp in its power of b ; it is kept because it is explicit, and the integer brackets below use it. (H_b is not Part II's harmonic number H_τ .)

Put $M_b = (b+1)2^{-b-2}$ and $V_b = L_b - M_b$. Integral comparison gives

$$L_b \leq (b/2)2^{-b}, \quad (b+1)3^{-b-2} \leq V_b \leq (b+4)3^{-b-2}.$$

Hence $M_b L_b / V_b \leq (9b/8)(3/4)^b \leq 729/512 < \tau$, so

$$d_b := \mu_\infty - \mu_b \geq \frac{L_b}{\tau(\tau + L_b)} > \frac{M_b}{\tau^2} = h(b).$$

The upper bounds above also give $0 < d_b/h(b) - 1 \leq 644\phi(b)$, where $\phi(x) = x^{3/2}(8/9)^x$. For $x \geq 13$, ϕ decreases, and for $x \geq 2$, $(\log h)'(x) < -1/3$. Every integer $b \leq b_0$ fails the threshold, proving its strict lower bracket. For $m = \lceil b_0 + 2000\phi(b_0) \rceil$,

$$\frac{d_m}{\varepsilon} \leq e^{-2000\phi(b_0)/3}(1 + 644\phi(b_0)) < 1,$$

proving the upper bracket. The constants are conservative for moderate caps. In particular, at $\varepsilon = h(n)$ for an integer $n \geq 3$, one has $b_0 = n$ but $b_\varepsilon \geq n + 1$. This supplies actual counterexamples to unconditional rounding by $\lceil b_0 \rceil$ at arbitrarily small tolerances.

Thus $b_\varepsilon = b_0 + O(1)$, and away from vanishingly small neighborhoods of integer transitions the threshold is the expected ceiling. Exactly near those transitions, an unconditional rounding prescription is not justified by an asymptotic equivalence. In particular this is an inverse for the cap of the limiting growth constant, not a cap selection guarantee for finite-length counts.

9 Numbers and reproducible checks

The following values are numerical quadratures of (1); the displayed digits are checks, not interval-certified enclosures.

b	T_b	μ_b
2	3.701101650408509482	0.270189823046234057
3	2.212067760911192191	0.452065717728321870
4	1.871791135820212368	0.534247641664251966
5	1.748278343303498299	0.571991298656956259
6	1.695001120651513698	0.589970111415399178
8	1.657736605427835731	0.603232139970701640
10	1.648371562426905637	0.606659337490447259
∞	1.644934066848226436	0.607927101854026629

In particular, for cap three (A317784),

$$a_n^{(3)} = n! (0.452065717728321870 \dots)^n \exp(O(n/\log n)),$$

where the constant is exactly the reciprocal integral, not the rounded decimal. The first ten counts, indexed from $n = 0$, are

$$\begin{aligned} b = 3 : & \quad 1, 1, 2, 5, 14, 47, 180, 773, 3701, 19488, \\ b = 4 : & \quad 1, 1, 2, 5, 15, 52, 210, 964, 4960, 28272. \end{aligned}$$

The accompanying source archive contains independent reproducibility routes:

- Direct generation of numerical words, unsorted raw-budget recursion, and grouped recursion agree for caps 3, 4 through $n = 9$.
- All 8,460 adjacent-budget state comparisons and 5,364 canonical raw/grouped state comparisons pass in the stated finite ranges.
- The actual suffix involution, its inverse, quotas, and ascent total pass 69,500 individual checks.
- Seeded tests of exact child-rank identities and the implementation of the barrier cover 82,801 transitions for $b = 3, \dots, 8$. Finite residual tests do not prove a uniform bound; Section 3 does.

- High-precision quadrature reproduces the constants, large-cap ratios, exact first variation, and numerical instances of both inverse models. It is not substituted for the analytic error estimates.

The commands, dependency list, test results, proof-source provenance, independent audits, and checksums are included. All mathematics needed for $b \geq 3$ is in this report; the separate cap-two document is identified by its file hash rather than duplicated. The PDF and T_EX source can be rebuilt with the supplied script. No external OEIS edit or publication is implied.

[write, 2 October 2026] In this repository the five checks and their saved results are shipped as `code/47-caps-*.py` and `data/47-caps-*.json`, the proof records and audits as `47-caps-*.md`, and the provenance record as `data/47-caps-provenance.json`. The delivered PDF and checksum manifest are not shipped, and the delivered T_EX source is the text of this Part. The README lists every file and gives a rerun recipe that leaves the shipped results untouched.

10 Literature and the remaining questions

Duncan and Steingrímsson [2] study pattern avoidance and growth rates for ascent sequences. Conway, Conway, Elvey Price and Guttmann [1] give the cap-two compacted recurrence and conjecture its factorial growth constant $8/(3\pi^2)$. The present arbitrary-budget involution generalizes that compaction mechanism. Their recurrence and conjectured constant are prior work, not a discovery claimed here.

The OEIS family entry A294220, retrieved on 1 October 2026, describes the maximum-letter-multiplicity array, identifies caps two and three as A202058 and A317784, and lists the finite values matching the checks above. Its retrieved formula section gives array relations and its program section gives exact enumeration. The standalone A317784 page could not be retrieved in this check; its identification here is verified through A294220's explicit cross-reference. No claim about A317784's current formula-section wording is made. Search-engine cache dates and the OEIS site footer do not certify an individual entry's revision date.

The report proves the statements presented here; it makes no exhaustive priority claim. The literature check is scoped, and conventional expert review and further literature screening remain appropriate before publication. The independent audits included with the package are mathematical checks of this argument, not journal peer review.

Several concrete questions remain:

1. Determine the subexponential factor of $a_n^{(b)}$ for fixed b : a prefactor, any power or stretched-exponential correction, and eventually a full coefficient expansion. The present error permits many possibilities.
2. Establish or refute the normalized ratio limit. A root limit by itself does not answer it.
3. Sharpen the rank barrier enough to quantify the true singular behavior of A_b near T_b , rather than only its radius and coefficient exponential scale.
4. Determine a joint regime $b = b(n)$ and the transition to unrestricted ascent sequences. The convergence of growth constants alone does not allow exchange of limits.
5. Extend the controlled large-cap first variation to further exponentially small sectors and their internal $1/b$ expansions, with uniform remainder estimates. Such a cap expansion is logically distinct from an all-orders expansion in n .

6. Find a direct combinatorial explanation for the truncated-exponential integral. A standard ascent-sequence/Fishburn bijection must not be assumed to preserve label multiplicity without proof.

[**write, 2 October 2026**] Question 5 is answered for every *fixed* number of sectors in Part II: the exact positive sectors $A_{r,b}$, the positive remainder after K sectors with its sharp size (Theorem 12.1), and an expansion of each fixed sector to every algebraic order in $1/b$ with rational coefficients (Theorem 13.1). Still open, as Part II says itself: estimates uniform in a growing sector index $r = r(b)$, the accumulation scale e^{-b} , optimal truncation and summability. For cap two, question 2 (the ratio limit) is answered in the sibling report `a202058-ascent-000-growth`; for every fixed $b \geq 3$ questions 1–4 and 6 remain open.

Part II

Fixed Taylor sectors for multiplicity cap growth constants

Source. Batch-77 manuscript 52, archive `oeis-fixed-sector-addendum.zip`, main file `fixed-sector-addendum.tex` (312 lines, 7-page PDF), dated 1 October 2026. Author line: “A reproducible large cap addendum”. No commit pin; it identifies source 47 by file hashes. An addendum to source 47 (Part I), printed in full, with labels prefixed `amc:sec:`; “the main report” and its citation `main` now refer to Part I. No symbol was renamed.

Abstract (as delivered). The cap integral has exact positive Taylor sectors and an exponentially sharp remainder after any fixed number of sectors. Each fixed sector has an algebraic expansion to arbitrary fixed order; its coefficients are rational and algorithmically computable. We give the leading constants, the first three corrections in sectors two and three, and a precise reciprocal corollary for the growth constant. This supplements the separately delivered bounded-multiplicity report [Part I of this report]: its combinatorial identification and inversion results are prerequisites, not reproved here. The statements are fixed-sector results. Finite algebraic truncations cannot generally resolve subsequent exponential sectors, and no uniform growing-sector or accumulation-scale theorem is asserted.

11 Objects and relationship to the main report

For integer $b \geq 2$, define

$$E_b(v) = \sum_{j=0}^b \frac{v^j}{j!}, \quad f(x) = \frac{\log x}{x-1}, \quad T_b = \int_0^\infty f(E_b(v)) dv, \quad (37)$$

with $f(1) = 1$. Set $\tau = \pi^2/6$, $m = b+1$, $R_b(v) = e^v - E_b(v)$, and $H_r = \sum_{j=1}^r 1/j$. The integral is finite for $b \geq 2$ because its integrand is $O_b(\log v/v^b)$ at infinity. Also $\int_0^\infty f(e^v) dv = \int_0^\infty v/(e^v - 1) dv = \tau$.

The main report, Part I of this report (Theorem 1.1), proves, for each fixed cap $b \geq 3$,

$$\lim_{n \rightarrow \infty} \left(\frac{a_n^{(b)}}{n!} \right)^{1/n} = \mu_b = T_b^{-1},$$

where $a_n^{(b)}$ counts ascent sequences in which every value occurs at most b times. Its cited cap-two theorem gives the same identification for $b = 2$. The present addendum refines the large- b analysis of T_b after that fixed-cap length limit. It does not strengthen the main report's error in n or give joint b, n asymptotics.

The main report established the exact first variation and a nonlinear bound $O(b^{5/2}(4/9)^b)$ (Part I, (30)). The theorem below sharpens that remainder to a full fixed-sector hierarchy, starting with $A_{2,b} \asymp b^{1/2}(4/9)^b$. The main report's controlled length inverse and rounding-safe cap inverse (Sections 6 and 8) remain the supplied inversion results; no sharper inverse is claimed in this addendum.

12 Exact positive sectors and the sharp remainder

Define, for $r \geq 1$,

$$a_r(x) = \frac{(-1)^r}{r!} f^{(r)}(x) = \int_0^1 \frac{t^r dt}{(1-t+tx)^{r+1}}, \quad A_{r,b} = \int_0^\infty a_r(e^v) R_b(v)^r dv. \quad (38)$$

These sectors are strictly positive. Define their primary exponential bases by

$$\beta_r = \left(\frac{r}{r+1} \right)^r. \quad (39)$$

They strictly decrease to e^{-1} ; in particular every fixed β_r is greater than e^{-1} .

Theorem 12.1. *For each integer $b \geq 2$, the following is an exact convergent identity:*

$$T_b - \tau = \sum_{r \geq 1} A_{r,b}. \quad (40)$$

For every fixed integer $K \geq 1$, as $b \rightarrow \infty$ through integers,

$$T_b - \tau = \sum_{r=1}^K A_{r,b} + \mathcal{R}_{K,b}, \quad (41)$$

$$0 < \mathcal{R}_{K,b} = O_K \left(b^{(2-K)/2} \beta_{K+1}^b \right), \quad (42)$$

$$\mathcal{R}_{K,b} = A_{K+1,b} \left[1 + O_K \left(b^{-1/2} \left(\frac{\beta_{K+2}}{\beta_{K+1}} \right)^b \right) \right]. \quad (43)$$

In particular $\mathcal{R}_{K,b} \sim A_{K+1,b}$, with a positive exact remainder. The estimates hold for fixed K ; the identity (40) does not license summing sector asymptotic formulas.

Proof. The integral representation $f(x) = \int_0^1 (1-t+tx)^{-1} dt$ gives (38). For $x = e^v$, $R = R_b(v)$, $E = x - R$, and $D = 1 - t + tx$, the finite geometric identity gives

$$f(E) - f(x) - \sum_{r=1}^K a_r(x) R^r = R^{K+1} \int_0^1 \frac{t^{K+1} dt}{D^{K+1}(D-tR)}. \quad (44)$$

Since $0 \leq tR/D \leq R/x < 1$, this lies between $a_{K+1}(x)R^{K+1}$ and $(x/E)a_{K+1}(x)R^{K+1}$. Expanding the infinite geometric series and using monotone convergence proves (40); in particular the sectors are integrable.

There is an absolute $c > 0$ such that $e^{-v}E_b(v) \geq c$ for $0 \leq v \leq b$. Indeed $e^{-v}E_b(v)$ is decreasing, while Stirling's bounds show that the $\lfloor \sqrt{b}/2 \rfloor + 1$ summands of $e^{-b}E_b(b)$ immediately below degree b are each at least a constant times $b^{-1/2}$. For those terms use

$$e^{-b} \frac{b^{b-j}}{(b-j)!} = \frac{e^{-b}b^b}{b!} \prod_{h=0}^{j-1} (1 - h/b), \quad \log(1 - u) \geq -2u \quad (0 \leq u \leq 1/2).$$

For sufficiently large b and $v \geq b$, $E_b(v) \geq v^b/b! \geq 2$ and $\log E_b(v) \leq v$. Thus

$$0 \leq \int_b^\infty [f(E_b(v)) - f(e^v)] dv \leq \frac{2b!b^{2-b}}{b-2} = O(b^{3/2}e^{-b}).$$

The integrated geometric bounds now give

$$A_{K+1,b} \leq \mathcal{R}_{K,b} \leq c^{-1}A_{K+1,b} + O(b^{3/2}e^{-b}). \quad (45)$$

Theorem 13.1 below, together with $\beta_{K+1} > e^{-1}$, proves (42). Applying the same bound with $K+1$ and using $\mathcal{R}_{K,b} = A_{K+1,b} + \mathcal{R}_{K+1,b}$ gives (43). Finally, the derivative of $-r \log(1 + 1/r)$ is negative, by $\log(1 + x) > x/(1 + x)$, proving the asserted ordering of the bases. $\square$

13 Every algebraic order within a fixed sector

Theorem 13.1. *For every fixed integer $r \geq 1$ and every fixed $N \geq 0$, there are rational coefficients $c_{r,j}$, with $c_{r,0} = 1$, such that*

$$A_{r,b} = C_r b^{(3-r)/2} \beta_r^b \left(\sum_{j=0}^N \frac{c_{r,j}}{b^j} + O_{r,N}(b^{-N-1}) \right), \quad C_r = \frac{r^{r+3/2}}{(r+1)^2} (2\pi)^{(1-r)/2}. \quad (46)$$

The first correction is

$$c_{r,1} = \frac{3-r}{2} - \frac{r^2(r+1)}{2} + \frac{23r}{12} + \frac{13}{12r} - \frac{r+1}{r} H_r. \quad (47)$$

In particular, the first three corrections in sectors two and three are

$$A_{2,b} = \frac{8}{9\sqrt{\pi}} \sqrt{b} \left(\frac{4}{9} \right)^b \left(1 - \frac{27}{8b} + \frac{5669}{128b^2} - \frac{962949}{1024b^3} + O(b^{-4}) \right), \quad (48)$$

$$A_{3,b} = \frac{81\sqrt{3}}{32\pi} \left(\frac{27}{64} \right)^b \left(1 - \frac{43}{3b} + \frac{33952}{81b^2} - \frac{42671116}{2187b^3} + O(b^{-4}) \right). \quad (49)$$

The first sector has the exact formula

$$A_{1,b} = (b+1)(\zeta(b+2) - 1). \quad (50)$$

Consequently its algebraic coefficients are $c_{1,1} = 1$ and $c_{1,j} = 0$ for $j \geq 2$; its smaller exponential terms are retained by (50).

13.1 Separate the derivative correction

Changing variables in (38) gives

$$a_r(e^v) = e^{-(r+1)v} \frac{v - \sum_{j=1}^r (1 - e^{-v})^j / j}{(1 - e^{-v})^{r+1}}, \quad (51)$$

with continuous value $1/(r+1)$ at $v = 0$. For $v \geq 1$ this equals $e^{-(r+1)v}(v - H_r) + O_r((1+v)e^{-(r+2)v})$. Define

$$J_{r,b} = \int_0^\infty (v - H_r) e^{-(r+1)v} R_b(v)^r dv. \quad (52)$$

The integrand of $J_{r,b}$ need not be positive. Coefficientwise, $R_b(v)^r \leq \sum_{k \geq rm} r^k v^k / k!$. Hence, for fixed $h > r$,

$$\int_0^\infty (1+v) e^{-hv} R_b(v)^r dv \leq \sum_{k \geq rm} \frac{r^k}{h^{k+1}} \left(1 + \frac{k+1}{h}\right) = O_{r,h}(m(r/h)^{rm}).$$

The bounded derivative factors on $[0, 1]$ and $R_b(v) \leq e/m!$ give

$$A_{r,b} = J_{r,b} + O_r \left(m \left(\frac{r}{r+2} \right)^{rm} + (m!)^{-r} \right). \quad (53)$$

The bases $(r/(r+2))^r$ are at most $1/3 < e^{-1}$, so these corrections are below every fixed primary sector. All constants here may depend on the fixed r .

For completeness, (50) can be evaluated without asymptotics. Put $F(k) = \int_0^\infty e^{-kv} R_b(v) dv = 1/[(k-1)k^m]$ for $k \geq 2$. Expanding $a_1(e^v) = \sum_{k \geq 2} ((k-1)v - 1)e^{-kv}$ and integrating gives

$$A_{1,b} = \sum_{k \geq 2} (-(k-1)F'(k) - F(k)) = m \sum_{k \geq 2} k^{-m-1}.$$

Interchanging the sum and integral is valid by absolute integrability: replacing each linear factor by $(k-1)v + 1$ gives an integrated summable bound. This also yields $J_{1,b} = m/2^{m+1}$.

[**write, 2 October 2026**] Since $A_{1,b}$ is Part I's L_b , this is a second proof of the first identity in (30). Part I (Section 7, "Evaluation of the linear term") integrates the double series in (h, k) term by term; this Part integrates against $F(k) = 1/((k-1)k^m)$ and differentiates in k .

13.2 Uniform tail expansion and saddle localization

Write $s = v/m$ and

$$R_b(ms) = \frac{(ms)^m}{m!} S_m(s), \quad S_m(s) = \sum_{\ell \geq 0} s^\ell \prod_{i=1}^\ell (1 + i/m)^{-1}. \quad (54)$$

For every fixed $d < 1$, to arbitrary fixed order and with any fixed number of derivatives,

$$S_m(s) = \sum_{j=0}^N \frac{\sigma_j(s)}{m^j} + O_{N,d}(m^{-N-1}), \quad \sigma_0(s) = \frac{1}{1-s}, \quad \sigma_j(s) = -\frac{s}{1-s} \sigma'_{j-1}(s), \quad (55)$$

uniformly on $0 \leq s \leq d$. To justify the remainder, expand $\prod_{i=1}^\ell (1 + iu)^{-1}$ at $u = 0$. Its absolute order- N remainder at $u = 1/m$ is bounded by $m^{-N-1}(\ell(\ell+1)/2)^{N+1}$. This follows by bounding its

alternating derivative at positive u by its value at zero. Summing the polynomial bound against s^ℓ proves uniformity, also after fixed derivatives. The exact differential equation $(1-s)S_m + (s/m)S'_m = 1$ then proves the recurrence in (55).

Substitution in (52) gives

$$J_{r,b} = \frac{m^{r+2}}{(m!)^r} \int_0^\infty (s - H_r/m) S_m(s)^r e^{m\phi_r(s)} ds, \quad \phi_r(s) = r \log s - (r+1)s. \quad (56)$$

Its unique nondegenerate saddle is $s_r = r/(r+1)$, with $\lambda_r = -\phi_r''(s_r) = (r+1)^2/r$. Choose fixed $0 < a < s_r < d < 1$. On $[0, a]$, use $S_m \leq (1-a)^{-1}$ and the phase gap below the maximum. On $[d, 1]$, the term ratios in (54) give $S_m \leq m+1$, and another phase gap defeats that polynomial. Finally the original integral over $v \geq m$ is $O_r(me^{-m})$, since $R_b(v) \leq e^v$. These contributions are exponentially smaller than the saddle term after Stirling's bounds are included.

On $[a, d]$, (55) and the interior Laplace expansion therefore apply to every fixed order. The leading term is

$$J_{r,b} \sim D_r m^{(3-r)/2} \beta_r^m, \quad D_r = r^{3/2} (r+1)^{r-2} (2\pi)^{(1-r)/2}.$$

Since $D_r \beta_r = C_r$, this proves the leading constant in (46). The same argument supplies a genuine remainder at each algebraic order, not merely a formal series. Together with (53), it establishes the all-orders assertion of Theorem 13.1.

13.3 A finite algorithm for all coefficients

The included author script uses the following exact operations to find $c_{r,0}, \dots, c_{r,N}$.

1. Set $z = m^{-1/2}$ and $s = s_r + uz$. Expand $(s - H_r z^2)(\sum_{j=0}^N \sigma_j(s) z^{2j})^r$ through degree $2N$ in z .
2. Multiply by the truncated exponential of the higher phase and Stirling terms

$$\sum_{k \geq 3} \frac{r(-1)^{k-1}}{k} (u/s_r)^k z^{k-2} - r \sum_{k \geq 1} \frac{B_{2k}}{2k(2k-1)} z^{4k-2},$$

retaining only the terms needed through z^{2N} .

3. Apply normalized Gaussian moments: $u^{2j} \mapsto (2j-1)!!/\lambda_r^j$, $u^{2j+1} \mapsto 0$. Divide by $s_r(1-s_r)^{-r}$ to obtain relative coefficients $d_{r,j}$ in powers of m^{-1} .
4. Put $p = (3-r)/2$. The coefficients in b are the coefficients of q^j in

$$\sum_{j=0}^N d_{r,j} q^j (1+q)^{p-j} \pmod{q^{N+1}}, \quad q = b^{-1}.$$

All these quantities are rational except for the already factored leading constant. In particular the first-order calculation gives

$$d_{r,1} = -\frac{r^2(r+1)}{2} + \frac{23r}{12} + \frac{13}{12r} - \frac{r+1}{r} H_r,$$

and $c_{r,1} = d_{r,1} + p$, proving (47). The explicit coefficients in (48)–(49) are independently reproduced as described in Section 15.

14 Reciprocation and the limits of exponential resolution

Corollary 14.1. *Use the combinatorial identification $\mu_b = T_b^{-1}$ from the main report (Theorem 1.1), and write $\mu_\infty = 1/\tau = 6/\pi^2$. For every fixed $K \geq 1$,*

$$\mu_\infty - \mu_b = \tau^{-2} \sum_{r=1}^K A_{r,b} + \tilde{\mathcal{R}}_{K,b}, \quad (57)$$

where

$$\tilde{\mathcal{R}}_{K,b} = \tau^{-2} A_{K+1,b} \left[1 + O_K \left(b^{-1/2} \left(\frac{\beta_{K+2}}{\beta_{K+1}} \right)^b \right) \right]. \quad (58)$$

Thus $\tilde{\mathcal{R}}_{K,b} > 0$ for all sufficiently large b for each fixed K . No all- b positivity of this reciprocal remainder is asserted.

Proof. Put $\delta_b = T_b - \tau$. The exact identity

$$\mu_\infty - \mu_b = \frac{\delta_b}{\tau^2} - \frac{\delta_b^2}{\tau^2(\tau + \delta_b)} \quad (59)$$

shows the scope of the linear sector transfer. Theorem 12.1 with $K = 1$ and (50) give $\delta_b = O(b2^{-b})$, so the second term is $O(b^2 4^{-b})$. For fixed K , this is exponentially smaller even than $A_{K+2,b}$, since $\beta_{K+2} > e^{-1} > 1/4$. Substituting (43) in (59) therefore gives (58). This proof uses no expansion with a growing number of sectors. $\square$

In particular,

$$\frac{6}{\pi^2} - \mu_b \sim \frac{9(b+1)}{\pi^4 2^b}.$$

Within each fixed primary sector, the coefficients of the deficit are $\tau^{-2} C_r c_{r,j}$. The exact reciprocal correction in (59) still exists on smaller exponential scales and must not be dropped in a purported complete transseries.

The essential scale warning. Theorems 12.1 and 14.1 require exact sectors, or approximations with errors controlled on the next exponential scale. If each $A_{r,b}$, $1 \leq r \leq K$, is replaced by only $N_r + 1$ terms of (46), one must add the error

$$O \left(\sum_{r=1}^K b^{(3-r)/2 - N_r - 1} \beta_r^b \right). \quad (60)$$

For any fixed truncation order, such a remainder in an earlier nonterminating sector generally exceeds all later exponential sectors. For example, subtracting only the displayed four terms of $A_{2,b}$ does not isolate $A_{3,b}$. Subtracting the exact sector does. The exact first sector is a useful exceptional simplification, not a cure for this issue in the later sectors.

The word “transseries” here describes an exact positive-sector decomposition with fixed-sector all-orders expansions. It does not assert optimal truncation, summability, uniform estimates for $r = r(b)$, a justified sum of all sector asymptotic expansions, or a theorem at the accumulation scale e^{-b} . Reciprocation and any subsequent inversion must preserve these distinctions.

[write, 2 October 2026] The word is used here in this narrow sense only. This Part constructs no transseries in the sense of the repository’s transseries volume ([Analysis/Transseries/](#)), no Borel sum and no Stokes data, which is why it stays beside Part I in this collection.

15 Independent checks and reproducibility

The fixed-sector derivation [6] received an independent PASS audit [7] before this addendum was assembled. That audit verified the exact positive remainder, localization and uniform tail expansion, every leading constant, and the coefficients through order b^{-3} . It is an audit of the integral analysis; the growth-constant identification remains the separate theorem in the main report.

The independent coefficient calculation expands an exact tail-index sum against independent geometric moments, rather than importing the author’s local Gaussian calculation. It reproduces the coefficient arrays for $r = 1, 2, 3, 4$ through degree three. A third included script checks $r = 1, 2, 3$ using Gamma-centered moments. These are computational cross-checks of the finite coefficients; the all-orders remainder is justified by the proof above.

For numerical independence, the audit expands

$$R_b(v)^r = \sum_{k=0}^r (-1)^k \binom{r}{k} e^{(r-k)v} E_b(v)^k.$$

If $E_b(v)^k = \sum_n d_{k,n} v^n / n!$, exact binomial convolution gives integers $d_{k,n}$ and the finite rational identity

$$J_{r,b} = \sum_{k=0}^r (-1)^k \binom{r}{k} \sum_n d_{k,n} \frac{(n+1) - (k+1)H_r}{(k+1)^{n+2}}. \quad (61)$$

Exact rational arithmetic avoids cancellation and uses no quadrature. Its normalized values for $r = 2, 3$ and $b = 50, 100, 200$ agree with the author’s saved values to at least 44 decimal places. The script also checks the derivative formula for $r = 1, \dots, 6$, $J_{1,b} = m/2^{m+1}$ for $b = 0, \dots, 30$, and three exact A_1 values. The full author quadrature run additionally covers $b = 500$ and exact A_2, A_3 values; it reproduces the saved JSON byte for byte in the recorded environment.

b	$J_{2,b}/(C_2 b^{1/2} \beta_2^b)$	$J_{3,b}/(C_3 \beta_3^b)$
50	0.945325812081121543	0.807327309466966331
100	0.969943924143225415	0.886287443120262425
200	0.984129422497738385	0.936943942102952663
500	0.993420049029619244	0.972871415163468681

These decimals illustrate convergence. The quadratures are high-precision numerical checks, not interval-certified bounds or substitutes for the proof.

Replay. The ZIP contains this PDF and editable TeX, the unmodified derivation and audit sources, the check scripts, their reference results, and provenance hashes. With Python 3, SymPy 1.14.0, and mpmath 1.3.0 installed, run `./verify.sh`; this writes fresh results to `checks/replay/`, checks all assertions, and compares them with the recorded outputs. Run `./build.sh` to rebuild the PDF with PDFLaTeX. Run `sha256sum -c SHA256SUMS` before changing supplied files. PDF metadata uses a fixed date; byte identity across different TeX installations is not promised. The package is offline-replayable after dependencies are installed.

[**write, 2 October 2026**] This paragraph describes the delivered archive, which survives in commit 096ee7b87. In this repository the scripts are `code/52-sectors-*`, the saved results `data/52-sectors-*`, the derivation and audit `52-sectors-fixed-taylor-sectors.md` and `52-sectors-fixed-sector-independent-audit.md`; `verify.sh` uses the delivered layout and

must not be run here. The README gives a rerun recipe. At intake (2 October 2026) the independent audit, the Gamma-moment check, the reciprocal check and the exact coefficients of the author script reproduced the saved results; the author’s 45-digit quadratures were not rerun.

16 Questions left open

1. How do the fixed-sector bounds behave as r grows with b ? The saddle $r/(r+1)$ approaches the endpoint 1, where the present compact-interval expansion loses uniformity. Identifying the correct joint regime is necessary before analyzing the accumulation near e^{-b} .
2. Can an optimally truncated or summable expansion for an earlier sector give a certified error below the next sector, and how do derivative exponentials and reciprocal products enter a complete smaller-scale description?
3. Can the refined exact-sector bounds yield a sharper, rounding-safe inverse for the cap at a prescribed deficit in μ_b ? Such an inverse needs explicit, appropriately scaled errors; finite coefficient substitution alone does not establish it.
4. For the original counting problem, can one determine fixed-cap prefactors or normalized ratio limits, and then obtain estimates uniform in a growing cap? None of the constant-level expansions here resolves those length-asymptotic questions.

References

- [1] A. R. Conway, M. Conway, A. Elvey Price and A. J. Guttmann, *Pattern-Avoiding Ascent Sequences of Length 3*, Electronic Journal of Combinatorics **29**(4) (2022), P4.25. doi:10.37236/11266. Published paper; arXiv:2111.01279.
- [2] P. Duncan and E. Steingrímsson, *Pattern avoidance in ascent sequences*, preprint (2011), arXiv:1109.3641.
- [3] OEIS Foundation Inc., A294220, maximum-letter-multiplicity array. Retrieved 1 October 2026.
- [4] OEIS Foundation Inc., A317784, cap-three column. Identification verified from A294220 on 1 October 2026.
- [5] *Factorial growth of ascent sequences avoiding 000*, companion proof and reproducible report, 1 October 2026. The separate deliverable is `a202058-report.pdf`; its SHA-256 and source hashes are recorded in this archive’s `provenance.json`. [Write note, 2 October 2026: batch-77 manuscript 37, printed in this repository as Part I of `SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/a202058-ascent-000-growth`; the record is shipped as `data/47-caps-provenance.json`.]
- [6] *Fixed Taylor sectors and their all-orders large-cap expansions*. Derivation dated 1 October 2026. Included as `sources/fixed-taylor-sectors.md`. [Shipped here as `52-sectors-fixed-taylor-sectors.md`.]
- [7] *Independent audit of the fixed Taylor sectors*. PASS audit dated 1 October 2026. Included as `sources/fixed-sector-independent-audit.md`; exact version recorded in `provenance.json`. [Shipped here as `52-sectors-fixed-sector-independent-audit.md` and `data/52-sectors-provenance.json`.]